

Packing arithmetic progressions with all differences in $[n]$: the leading constant is $4/3$

Conjecture 1 of Alon, Dębski, Grytczuk and Przybyło

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Machine-generated note

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Abstract

Let $A_d(n) = \{d, 2d, \dots, \lfloor n/d \rfloor d\}$, and let $m(n)$ be the least cardinality of an integer interval containing pairwise disjoint translates of all $A_d(n)$, $1 \leq d \leq n$. We prove

$$m(n) \sim \frac{4}{3} \frac{n^{3/2}}{\log n},$$

settling Conjecture 1 of Alon, Dębski, Grytczuk and Przybyło. The prime-difference phase-selection mechanism overlaps with independent prior work of Hou, Liu and Zhao; the additional bundle coloring and Bohr-set clustering incorporate all composite differences.

This note was produced by AI within the [Graph Theory AI](#) project: the argument was found by a language model and audited by an adversarial LLM referee, and it has not been checked by a human mathematician. [Section 10](#) records the full provenance.

1 Introduction

Translations are by integers, and packing length means interval cardinality. Replacing cardinality by diameter changes the parameter by at most one. Put

$$A_d(n) = \{d, 2d, \dots, \lfloor n/d \rfloor d\}, \quad x = \sqrt{n}, \quad T_n = \frac{n^{3/2}}{\log n}. \quad (1.1)$$

For $D \subseteq [n]$, write $m_D(n)$ for the least length of an interval containing disjoint translates of the $A_d(n)$ for $d \in D$; thus $m(n) = m_{[n]}(n)$.

Alon, Dębski, Grytczuk and Przybyło proved lower and upper leading constants $4/3$ and $5/3$ and proposed the following exact asymptotic [[1](#), Conjecture 1].

Conjecture 1.1 ([[1](#), Conjecture 1]).

$$m(n) \sim \frac{4}{3} \cdot \frac{n^{3/2}}{\ln n}.$$

Shortly before the machine run recorded here, Hou, Liu and Zhao proved the corresponding $4/3$ asymptotic for the subfamily of prime differences $p \leq \sqrt{n}$ [[2](#), Theorem 1.1]. Their regular-prime-block argument and the generic phase control below address the same obstacle by related methods. They explicitly leave the full difference set open because differences $p, 2p, \dots$ are not pairwise coprime. The new part of this note is the mechanism that compresses every bundle $\{A_{ap}(n)\}_a$ into $o(p)$ residue classes, keeps those classes close to one selected CRT phase, and covers the remaining indices at lower-order cost.

Theorem 1.2 (Main theorem). *As $n \rightarrow \infty$,*

$$m(n) = \left(\frac{4}{3} + o(1) \right) \frac{n^{3/2}}{\log n}.$$

Idea of proof. The prime-indexed progressions give the $4/3$ lower bound by a pairwise separation inequality. For the upper bound, color all progressions whose differences are multiples of one prime into $o(p)$ classes, each class already disjoint without shifts. Fixed-size generic blocks of primes admit simultaneous control of every pairwise CRT intersection phase. A centered Bohr set supplies enough nearby residue perturbations to place all color classes while retaining that phase control. Summing these blocks yields the optimal integral $2 \int_0^1 (1 - u^2) du = 4/3$ for the main range. A divisor cover, a folding lemma, and a greedy estimate put every remaining difference into $o(T_n)$ plus $O(T_n/C)$ additional space. [\[full proof\]](#)

2 The prime lower bound

Lemma 2.1 (Pairwise separation). *Let $p, q \leq \sqrt{n}$ be distinct primes. If $t + A_p(n)$ and $u + A_q(n)$ are disjoint and $u \geq t$, then*

$$u - t > \left\lfloor \frac{n}{p} \right\rfloor p - pq. \quad (2.1)$$

Idea of proof. If the difference were smaller, choose $j \in \{1, \dots, p\}$ so that $u - t + jq$ is divisible by p . The resulting common value lies within both finite progressions because $pq \leq n$. [\[full proof\]](#)

Proposition 2.2.

$$m(n) \geq \left(\frac{4}{3} - o(1) \right) T_n. \quad (2.2)$$

Idea of proof. Order the translation bases of all prime-indexed progressions and sum [Lemma 2.1](#) over consecutive bases. The saving is bounded by $\sum_{p \leq x} p^2$. The prime number theorem and partial summation give $n\pi(x) \sim 2T_n$ and $\sum_{p \leq x} p^2 \sim \frac{2}{3}T_n$. [\[full proof\]](#)

3 Packing all multiples of one modulus

For $q \geq 1$, define the bundle

$$\mathcal{B}_q(n) = \{A_{qa}(n) : 1 \leq a \leq \lfloor n/q \rfloor\}.$$

Lemma 3.1 (Bundle coloring). *There is an absolute constant K_0 such that, for every $N \geq 1$, the family*

$$\{A_a(N) : 1 \leq a \leq N\}$$

can be partitioned into at most

$$R(N) \leq K_0 N^{3/5} \quad (3.1)$$

classes, each consisting of pairwise disjoint unshifted progressions.

Idea of proof. Give every $a \leq \sqrt{N}$ its own color. For $a > \sqrt{N}$, two unshifted progressions intersect only if the other index divides some $ka \leq N$. The intersection graph therefore has degree at most $\sqrt{N} \max_{t \leq N} \tau(t) = O(N^{3/5})$, using the elementary fixed-exponent divisor bound; greedy coloring finishes. [\[full proof\]](#)

Referee repair (divisor bound). The full proof records a finite, explicit product bounding the contribution of primes below 2^{10} . This makes $\tau(t) \leq Kt^{1/10}$ rigorous, although the constant is enormous and only the asymptotic $R(N) = o(\sqrt{N})$ is used.

Corollary 3.2. *The bundle $\mathcal{B}_q(n)$ can be packed in an interval of cardinality at most*

$$(n+q) \left\lceil \frac{R(\lfloor n/q \rfloor)}{q} \right\rceil. \quad (3.2)$$

In particular, if $p \asymp \sqrt{n}$ is prime, then

$$R(\lfloor n/p \rfloor) = O(n^{3/10}) = o(p). \quad (3.3)$$

Idea of proof. Since $A_{qa}(n) = qA_a(\lfloor n/q \rfloor)$, put at most q color classes in one batch and assign them distinct residues modulo q . Concatenate the batches. [\[full proof\]](#)

4 Generic prime blocks

Let $\mathbb{T} = \mathbb{R}/\mathbb{Z}$. For distinct primes p_i, p_j and residues $r_i \bmod p_i, r_j \bmod p_j$, define

$$Z_{ij}(r_i, r_j) = \frac{\text{CRT}_{p_i, p_j}(r_i, r_j)}{p_i p_j} \pmod{1}, \quad (4.1)$$

where the representative lies in $\{0, \dots, p_i p_j - 1\}$.

Lemma 4.1 (Generic prime blocks). *Fix $k \geq 2$, $0 < a < b$, and $0 < \eta < 1$. There is $L = L(k, a, b, \eta)$ such that, for all sufficiently large X , every set of primes in $[aX, bX]$ can be partitioned into k -element blocks and at most L exceptional primes, and every block $p_1, \dots, p_k$ has the following property: for arbitrary open arcs $J_{ij} \subseteq \mathbb{T}$, $i < j$, each of length η , there are residues $r_i \bmod p_i$ with*

$$Z_{ij}(r_i, r_j) \in J_{ij} \quad (i < j). \quad (4.2)$$

Idea of proof. For uniform independent r_i , the Fourier coefficient indexed by $h = (h_{ij})$ vanishes unless one congruence holds at every p_i . A smooth bump supported strictly inside each prescribed arc reduces positivity to finitely many frequencies $\|h\|_\infty \leq H$. For each nonzero such h , fixing $k-1$ primes leaves only $O(1)$ possibilities for the last prime, so only $O(M^{k-1})$ of the M^k ordered blocks are bad. Repeatedly remove good blocks. [\[full proof\]](#)

Referee repair (open arcs). The Fourier bump is chosen with support in an arc of length strictly less than η (for example $\eta/2$), then translated inside each given open η -arc. Positivity therefore yields membership in the prescribed arcs, including the endpoint issue omitted from the original writeup.

5 A large centred Bohr set

Lemma 5.1. *Fix $d \geq 1$ and $\rho > 0$. There is $c = c(d, \rho) > 0$ such that, for every $q \geq 2$ and every $c_1, \dots, c_d \in \mathbb{Z}$, at least cq residues $s \bmod q$ satisfy*

$$\left\| \frac{c_j s}{q} \right\|_{\mathbb{T}} < \rho \quad (1 \leq j \leq d). \quad (5.1)$$

Idea of proof. Partition the d -torus into M^d boxes of side less than ρ . One box contains q/M^d of the points $(c_1 s/q, \dots, c_d s/q)$; subtract one representative from the others. [\[full proof\]](#)

6 Packing the main prime range

Fix $0 < \epsilon < 1 < C$, and let $\mathcal{G}_{\epsilon,C}(n)$ contain all $A_d(n)$ for which d has a prime divisor in $[\epsilon x, Cx]$.

Proposition 6.1 (Main-range packing).

$$\limsup_{n \rightarrow \infty} \frac{m(\mathcal{G}_{\epsilon,C}(n))}{T_n} \leq 2 \int_{\epsilon}^C (1 - u^2)_+ du. \quad (6.1)$$

Consequently, for $C > 1$,

$$m(\mathcal{G}_{\epsilon,C}(n)) \leq \left(\frac{4}{3} - 2\epsilon + \frac{2}{3}\epsilon^3 + o(1) \right) T_n. \quad (6.2)$$

Idea of proof. Pack labeled copies of all bundles $\mathcal{B}_p(n)$, deleting duplicates afterward. In a small prime bin, group primes into generic k -blocks. Give consecutive bundles host intervals separated by approximately $n(1 - u^2)$; this makes the projected overlap a proper forbidden arc modulo every $p_i p_j$. [Lemma 4.1](#) chooses central CRT phases outside all forbidden arcs, while [Lemma 5.1](#) supplies one nearby perturbation for every color from [Lemma 3.1](#). Thus all composite-indexed progressions in a block are disjoint. Summing blocks, refining the bins, then sending the fixed parameters $\nu \downarrow 0$ and $k \rightarrow \infty$ yields the integral. [\[full proof\]](#)

7 Two tools for the large-prime tail

Lemma 7.1 (Folding a scaled packing). *Suppose a family of finite integer sets, each of diameter at most N , has a packing in an interval of cardinality M . For an integer $a \geq 1$, the family obtained by multiplying every set by a has a packing of cardinality at most*

$$M + aN + a. \quad (7.1)$$

Idea of proof. Divide the original interval into a strips of width $\lceil M/a \rceil$. Translate each strip to the origin, dilate by a , and label it by a distinct residue modulo a . [\[full proof\]](#)

Lemma 7.2 (Large prime indices). *There is an absolute constant B such that, whenever $P \geq 2\sqrt{N}$,*

$$m_{\{p \text{ prime}: P < p \leq N\}}(N) \leq B \left(\frac{N^2}{P \log P} + N \log(2N) \right). \quad (7.2)$$

Idea of proof. In a dyadic bin $(Q, 2Q]$, greedily pack $K = \lfloor Q^2/(2N) \rfloor$ progressions into $[1, 2N]$. The already occupied points forbid fewer than $Q/2 < p$ residues for the next prime p . Grouping and summing the dyadic bins gives (7.2). [\[full proof\]](#)

8 Covering every difference

Proposition 8.1. *For every fixed $0 < \epsilon < 1$ and $C > 2$,*

$$\limsup_{n \rightarrow \infty} \frac{m(n)}{T_n} \leq \frac{4}{3} + \frac{2}{3}\epsilon^3 + \frac{B_0}{C} \quad (8.1)$$

for an absolute constant B_0 .

Idea of proof. Put $Y = \lceil n^{1/5} \rceil$. Besides the main family, use bundles with divisors $q \in [Y, Y^2]$ or primes $q \in [Y, \epsilon x]$, pack the $d < Y$ progressions separately, and treat each remaining $d = ap$ with $a < Y$ and prime $p > Cx$ by [Lemmas 7.1](#) and [7.2](#). Every $d \leq n$ has one of these forms. The small bundles cost $(2\epsilon + o(1))T_n$, canceling the missing 2ϵ in (6.2); the large-prime tails cost at most $(B_0/C + o(1))T_n$. [\[full proof\]](#)

9 Remarks

Remark 9.1 (Relation to prior work). The prime-range half of the proof, including the mechanism producing the constant $4/3$, overlaps with Hou–Liu–Zhao [2], posted on 7 September 2026, nine days before the machine attack began. Their regular-block proof uses transference, while [Lemma 4.1](#) uses Fourier truncation and counting. The full theorem here is not contained in that preprint: its concluding discussion leaves arbitrary differences open because of their common divisors. The bundle coloring, Bohr clustering, folding, and complete cover in [Sections 3, 5, 7 and 8](#) supply precisely that extension.

Remark 9.2 (Fragility of genericity). Genericity in [Lemma 4.1](#) is not automatic for every prime block. For every triple $p_0 < p_1 < p_2$, the vector

$$(h_{01}, h_{02}, h_{12}) = (p_1 - p_0, -(p_2 - p_0), p_2 - p_1)$$

satisfies all three Fourier congruences. The lemma works because its cutoff $H(k, \eta)$ is fixed while $X \rightarrow \infty$; only $O(M^{k-1})$ tuples have a relation below that fixed cutoff.

Remark 9.3 (Qualitative nature). The constants in the divisor estimate, Fourier cutoff, exceptional-block count, and Bohr density are not effective at a useful scale. The referee found the normalized prime lower-bound expression equal to 1.639, 1.605, 1.572, 1.545, 1.522 for $n = 10^6, \dots, 10^{10}$, still far from $4/3$. The theorem asserts no convergence rate.

Remark 9.4 (Computational audit). At $n = 10^8$, a literal three-bundle instance with primes 15013, 15511, 15991 placed all 172 850 points without collision in a span 119 973 288, below the predicted 120 020 001; all three host windows overlapped and composite differences were included. The referee also checked the bundle packings, 200 random folding instances, two large-prime greedy instances, and the complete index cover for $n = 10^4, 10^5, 10^6$, with no uncovered index. The limiting order of parameters was verified only analytically.

Remark 9.5 (Other recent progress). Mao, Wang, Wei and Yang [3] settled another conjecture from the source paper, the fixed-cardinality asymptotic $M_k(n) = (1 + o(1))nk$ for fixed n . It is independent of the result proved here.

10 Provenance

The mathematics in this note was produced without human intervention, in three stages. (1) The proof was found by gpt-6-astra, reasoning effort max, in a single pass begun on 2026-09-16, catalog id 2603.02786__00, leg attacks_retry. This was a second attempt, with the earlier gpt-5.6-sol attempt and its partial verdict supplied as context. That attempt proved only prime-subfamily results and explicitly failed to incorporate composite differences; the retry replaced its path restriction with simultaneous CRT control and bundle coloring.

(2) An adversarial referee agent (claude-opus-5) audited the writeup on 2026-09-17 and returned MINOR GAPS with medium confidence. It re-derived all 24 proof steps, found the prior prime-only result [2], and ran the symbolic, number-theoretic, block-packing, folding, greedy, and index-cover checks summarized above.

(3) A rewrite was started by claude-fable-5-1 on 2026-09-18 but stopped at a compilable skeleton when that session hit its rate limit. The interrupted transcript and skeleton were recovered, and Codex completed the present note on 2026-09-18, with proof ideas in the main text and full proofs in [Appendix A](#). Relative to the original writeup it makes the following declared changes: it cites and distinguishes Hou–Liu–Zhao’s prior prime-only theorem; adds the related fixed-cardinality result [3]; expands the fixed-exponent divisor bound to include its finite small-prime constant; chooses the smooth Fourier bump with support strictly shorter than the prescribed open arcs; highlights the universal large-frequency relation for prime triples; spells out deletion of duplicate labelled copies; and records the scope and computational limitations. No other mathematical content was changed. **No human mathematician has checked this argument.**

A Full proofs

Lemma 2.1 (Pairwise separation). *Let $p, q \leq \sqrt{n}$ be distinct primes. If $t + A_p(n)$ and $u + A_q(n)$ are disjoint and $u \geq t$, then*

$$u - t > \left\lfloor \frac{n}{p} \right\rfloor p - pq. \quad (2.1)$$

Proof. Put $\delta = u - t$ and $r_p = \lfloor n/p \rfloor$. If $0 \leq \delta \leq r_p p - pq$, choose $j \in \{1, \dots, p\}$ such that $\delta + jq \equiv 0 \pmod{p}$, possible because q is invertible modulo p . Let $i = (\delta + jq)/p$. Then $i \geq 1$ and

$$ip = \delta + jq \leq r_p p - pq + pq = r_p p,$$

so $i \leq r_p$. Also $j \leq p \leq \lfloor n/q \rfloor$ because $pq \leq n$. Thus $t + ip = u + jq$ lies in both translates, a contradiction. $\square$

Proposition 2.2.

$$m(n) \geq \left(\frac{4}{3} - o(1) \right) T_n. \quad (2.2)$$

Proof. Pack all $A_p(n)$ with $p \leq x$, and order their bases as $t_{p_1} \leq \dots \leq t_{p_s}$. Summing Lemma 2.1 gives

$$t_{p_s} - t_{p_1} \geq (s-1)n - \sum_{i=1}^{s-1} p_i p_{i+1} - O(n),$$

where the accumulated rounding error is $O(\pi(x)x) = O(n)$. Moreover,

$$\sum_{i=1}^{s-1} p_i p_{i+1} \leq \frac{1}{2} \sum_{i=1}^{s-1} (p_i^2 + p_{i+1}^2) \leq \sum_{p \leq x} p^2.$$

Bases of progressions in one interval of cardinality M differ by at most $M + x$, so

$$M \geq n\pi(x) - \sum_{p \leq x} p^2 - O(n).$$

The prime number theorem and partial summation [4, Chs. XXII–XXIII] give

$$\pi(x) \sim \frac{x}{\log x}, \quad \sum_{p \leq x} p^2 \sim \frac{x^3}{3 \log x}.$$

Since $\log n = 2 \log x$, these terms are $2T_n$ and $\frac{2}{3}T_n$, proving (2.2). $\square$

Lemma 3.1 (Bundle coloring). *There is an absolute constant K_0 such that, for every $N \geq 1$, the family*

$$\{A_a(N) : 1 \leq a \leq N\}$$

can be partitioned into at most

$$R(N) \leq K_0 N^{3/5} \quad (3.1)$$

classes, each consisting of pairwise disjoint unshifted progressions.

Proof. Give every $a \leq \sqrt{N}$ its own color. For $a > \sqrt{N}$, form the intersection graph of the unshifted sets $A_a(N)$. If $A_a(N)$ meets $A_b(N)$, then $\text{lcm}(a, b) = ka \leq N$ for some $k \leq N/a < \sqrt{N}$, and $b \mid ka$. Hence the degree of a is at most

$$\sum_{k \leq N/a} \tau(ka) \leq \sqrt{N} \max_{t \leq N} \tau(t). \quad (\text{A.1})$$

We record the fixed-exponent divisor bound. Write $t = \prod_p p^{e_p}$. For $p \geq 2^{10}$,

$$e_p + 1 \leq 2^{e_p} \leq p^{e_p/10}.$$

For each of the finitely many primes $p < 2^{10}$, the constant

$$C_p = \sup_{e \geq 0} \frac{e + 1}{p^{e/10}}$$

is finite because the denominator grows exponentially. Thus

$$\tau(t) = \prod_p (e_p + 1) \leq \left(\prod_{p < 2^{10}} C_p \right) t^{1/10} = K t^{1/10}.$$

Equation (A.1) gives maximum degree at most $KN^{3/5}$. Greedy coloring, together with the at most $\sqrt{N}$ private colors, proves the result after increasing the absolute constant K_0 . $\square$

Corollary 3.2. *The bundle $\mathcal{B}_q(n)$ can be packed in an interval of cardinality at most*

$$(n + q) \left\lceil \frac{R(\lfloor n/q \rfloor)}{q} \right\rceil. \quad (3.2)$$

In particular, if $p \asymp \sqrt{n}$ is prime, then

$$R(\lfloor n/p \rfloor) = O(n^{3/10}) = o(p). \quad (3.3)$$

Proof. The identity

$$A_{qa}(n) = qA_a(\lfloor n/q \rfloor)$$

follows from $\lfloor \lfloor n/q \rfloor / a \rfloor = \lfloor n/(qa) \rfloor$. Color the right-hand progressions using Lemma 3.1 and divide the colors into batches of at most q . Within one batch assign a distinct residue $r \in \{0, \dots, q-1\}$ to every color and translate all progressions of that color by r . Same-color sets remain disjoint, while different colors are separated modulo q . Each batch lies in an interval of length at most $n + q$; concatenating proves (3.2). If $p \asymp \sqrt{n}$, then $R(\lfloor n/p \rfloor) = O((n/p)^{3/5}) = O(n^{3/10}) = o(p)$. $\square$

Lemma 4.1 (Generic prime blocks). *Fix $k \geq 2$, $0 < a < b$, and $0 < \eta < 1$. There is $L = L(k, a, b, \eta)$ such that, for all sufficiently large X , every set of primes in $[aX, bX]$ can be partitioned into k -element blocks and at most L exceptional primes, and every block $p_1, \dots, p_k$ has the following property: for arbitrary open arcs $J_{ij} \subseteq \mathbb{T}$, $i < j$, each of length η , there are residues $r_i \bmod p_i$ with*

$$Z_{ij}(r_i, r_j) \in J_{ij} \quad (i < j). \quad (4.2)$$

Proof. Put $E = \binom{k}{2}$. For $i \neq j$, let $\overline{p_j^{(p_i)}}$ be the inverse of p_j modulo p_i . The CRT formula gives

$$Z_{ij}(r_i, r_j) = \frac{\overline{p_j^{(p_i)}} r_i}{p_i} \frac{\overline{p_i^{(p_j)}} r_j}{p_j} \pmod{1}. \quad (A.2)$$

Choose the r_i independently and uniformly. For $h = (h_{ij})_{i < j} \in \mathbb{Z}^E$, write $h_{ji} = h_{ij}$. Independence gives

$$\mathbb{E} \exp \left(2\pi i \sum_{i < j} h_{ij} Z_{ij} \right) = \prod_{i=1}^k \mathbf{1} \left[\sum_{j \neq i} h_{ij} \overline{p_j^{(p_i)}} \equiv 0 \pmod{p_i} \right]. \quad (A.3)$$

Choose a nonnegative smooth periodic function f with positive integral μ , supported in an arc of length $\eta/2$. For each prescribed open η -arc J_{ij} , translate this support into J_{ij} and call its center c_{ij} . The product

$$F(z) = \prod_{i < j} f(z_{ij} - c_{ij})$$

has constant Fourier coefficient μ^E . Its Fourier series is absolutely convergent, and the absolute coefficient sum outside $\|h\|_\infty \leq H$ tends to zero uniformly in the translations. Fix $H = H(k, \eta)$ so the tail is below $\mu^E/2$. If every nonzero coefficient in that box vanishes under (A.3), then $\mathbb{E}F(Z) \geq \mu^E/2 > 0$, so some residue vector has all phases inside the required arcs.

Call an ordered prime tuple bad if a nonzero h with $\|h\|_\infty \leq H$ makes every congruence in (A.3) hold. Fix h and choose i, j with $h_{ij} \neq 0$. After all primes but p_j are fixed, the congruence at p_i is

$$h_{ij} \overline{p_j}^{(p_i)} \equiv -c \pmod{p_i}. \quad (\text{A.4})$$

For $p_i > H$, if $c = 0$ there is no solution; otherwise (A.4) pins p_j to one residue class modulo p_i . Since $p_i \geq aX$ and $p_j \in [aX, bX]$, only $O_{a,b}(1)$ integers qualify. Summing over the finitely many h shows that a set of M primes has $O_{k,a,b,\eta}(M^{k-1})$ bad ordered tuples. For M above a fixed threshold this is less than $(M)_k$, so a good block exists. Repeatedly remove one until only boundedly many primes remain. $\square$

Lemma 5.1. Fix $d \geq 1$ and $\rho > 0$. There is $c = c(d, \rho) > 0$ such that, for every $q \geq 2$ and every $c_1, \dots, c_d \in \mathbb{Z}$, at least cq residues $s \pmod{q}$ satisfy

$$\left\| \frac{c_j s}{q} \right\|_{\mathbb{T}} < \rho \quad (1 \leq j \leq d). \quad (5.1)$$

Proof. Choose an integer $M > 1/\rho$. Partition $[0, 1)^d$ into M^d congruent boxes and consider

$$\left(\frac{c_1 s}{q}, \dots, \frac{c_d s}{q} \right) \pmod{1}, \quad s \pmod{q}.$$

Some box contains at least q/M^d residues. Fix one, s_0 , and subtract it from all the others. The differences remain distinct modulo q and satisfy (5.1). Take $c = M^{-d}$. $\square$

Proposition 6.1 (Main-range packing).

$$\limsup_{n \rightarrow \infty} \frac{m(\mathcal{G}_{\epsilon, C}(n))}{T_n} \leq 2 \int_{\epsilon}^C (1 - u^2)_+ du. \quad (6.1)$$

Consequently, for $C > 1$,

$$m(\mathcal{G}_{\epsilon, C}(n)) \leq \left(\frac{4}{3} - 2\epsilon + \frac{2}{3}\epsilon^3 + o(1) \right) T_n. \quad (6.2)$$

Proof. It suffices to pack labeled copies of every bundle $\mathcal{B}_p(n)$ for primes $p \in [\epsilon x, Cx]$: if an index has two such prime factors, delete all but one packed copy afterward.

Fix $k \geq 2$, $0 < \nu < \epsilon^2/4$, and put

$$h(u) = (1 - u^2)_+, \quad \rho = \frac{\nu}{100C^2}.$$

Consider a generic block $p_1, \dots, p_k \in [ux, vx]$. Set

$$g = \lceil n(h(u) + \nu) \rceil, \quad W = n + \lceil Cx \rceil + 1, \quad S_i = (i - 1)g,$$

and use host intervals $I_i = [S_i, S_i + W - 1]$. If I_i, I_j overlap, then for large n ,

$$W - g \leq (\min(u^2, 1) - \nu/2)n \leq p_i p_j - \nu n/2. \quad (\text{A.5})$$

After projection modulo $p_i p_j$, the overlap is a forbidden arc whose complement has length at least $\nu/(2C^2) = 50\rho$. Choose an open ρ -arc J_{ij} in the complement, at circular distance at least 4ρ from the forbidden arc. If the hosts are disjoint, choose any arc. [Lemma 4.1](#) provides residues $r_i \bmod p_i$ with $Z_{ij}(r_i, r_j) \in J_{ij}$.

Put $N_i = \lfloor n/p_i \rfloor$ and color the $A_a(N_i)$ by [Lemma 3.1](#). By (3.3), the number of colors is $o(p_i)$. Apply [Lemma 5.1](#) with coefficients $\overline{p_j^{(p_i)}}$, $j \neq i$. It gives a positive proportion of residues $s \bmod p_i$ satisfying

$$\left\| \frac{\overline{p_j^{(p_i)}} s}{p_i} \right\|_{\mathbb{T}} < \rho \quad (j \neq i). \quad (\text{A.6})$$

For large n there are enough distinct such residues $s_{i,c}$ for all colors c . Choose $t_{i,c} \in \{0, \dots, p_i - 1\}$ so that

$$S_i + t_{i,c} \equiv r_i + s_{i,c} \pmod{p_i},$$

and translate every $A_{ap_i}(n)$ of color c by $S_i + t_{i,c}$. These sets lie in I_i . Same-color progressions are disjoint by the coloring, and different colors occupy different residues modulo p_i .

For two bundles, the intersection phase is

$$\begin{aligned} Z_{ij}(r_i + s_{i,c}, r_j + s_{j,c'}) &= Z_{ij}(r_i, r_j) \\ &\quad + \frac{\overline{p_j^{(p_i)}} s_{i,c}}{p_i} + \frac{\overline{p_i^{(p_j)}} s_{j,c'}}{p_j} \pmod{1}. \end{aligned} \quad (\text{A.7})$$

It moves by less than 2ρ from J_{ij} and remains outside the forbidden projection. Hence no point lies in progressions from two bundles. The block occupies at most

$$(k-1)g + W. \quad (\text{A.8})$$

Partition $[\epsilon, C]$ into fixed bins $\epsilon = u_0 < \dots < u_J = C$. Apply [Lemma 4.1](#) in each bin. An exceptional prime's entire bundle fits in length W because eventually $R(N_i) \leq p_i$, so all exceptions cost $O(n) = o(T_n)$. With

$$M_j = \#\{p : u_{j-1}x \leq p < u_j x\}, \quad g_j = \lceil n(h(u_{j-1}) + \nu) \rceil,$$

concatenating (A.8) gives

$$m(\mathcal{G}_{\epsilon,C}(n)) \leq \sum_{j=1}^J M_j \left(g_j + \frac{W}{k} \right) + O(n).$$

The prime number theorem yields

$$M_j = (u_j - u_{j-1} + o(1)) \frac{x}{\log x}.$$

Divide by T_n , let $n \rightarrow \infty$ first, then refine the partition, let $\nu \downarrow 0$, and finally $k \rightarrow \infty$. All auxiliary parameters were fixed before the n limit, and the upper Riemann sums converge to

$$2 \int_{\epsilon}^C (1 - u^2)_+ du.$$

For $C > 1$ this integral is $\frac{4}{3} - 2\epsilon + \frac{2}{3}\epsilon^3$. □

Lemma 7.1 (Folding a scaled packing). *Suppose a family of finite integer sets, each of diameter at most N , has a packing in an interval of cardinality M . For an integer $a \geq 1$, the family obtained by multiplying every set by a has a packing of cardinality at most*

$$M + aN + a. \quad (7.1)$$

Proof. Translate the original packing into $[0, M - 1]$ and put $H = \lceil M/a \rceil$. Assign a set B to the strip $jH \leq \min B < (j + 1)H$, $0 \leq j < a$, and replace it by

$$a(B - jH) + j.$$

Sets from one strip remain disjoint under the affine injection; sets from different strips have distinct residues modulo a . Because each diameter is at most N , all images lie in an interval of length at most

$$aH + aN \leq M + aN + a.$$

□

Lemma 7.2 (Large prime indices). *There is an absolute constant B such that, whenever $P \geq 2\sqrt{N}$,*

$$m_{\{p \text{ prime: } P < p \leq N\}}(N) \leq B \left(\frac{N^2}{P \log P} + N \log(2N) \right). \quad (7.2)$$

Proof. Use dyadic bins $(Q, 2Q]$, starting at $Q = P$, and put

$$K = \left\lfloor \frac{Q^2}{2N} \right\rfloor \geq \frac{Q^2}{4N},$$

where the inequality uses $Q \geq 2\sqrt{N}$. Any group of at most K prime-indexed progressions from the bin fits in $[1, 2N]$. Indeed, insert them successively. For a new prime p , every occupied point forbids at most one shift residue modulo p , while previous progressions contain fewer than

$$K \frac{N}{Q} \leq \frac{Q}{2} < p$$

points. Some residue remains.

The total length for this bin is at most

$$2N \left(\frac{4N\pi(2Q)}{Q^2} + 1 \right).$$

Using $\pi(t) = O(t/\log t)$ and

$$\sum_{Q=P, 2P, \dots} \frac{1}{Q \log Q} \leq \frac{2}{P \log P},$$

while there are $O(\log(2N))$ bins, gives (7.2) after enlarging an absolute constant B . □

Proposition 8.1. *For every fixed $0 < \epsilon < 1$ and $C > 2$,*

$$\limsup_{n \rightarrow \infty} \frac{m(n)}{T_n} \leq \frac{4}{3} + \frac{2}{3}\epsilon^3 + \frac{B_0}{C} \quad (8.1)$$

for an absolute constant B_0 .

Proof. Set $Y = \lceil n^{1/5} \rceil$, so $Y^2 < \epsilon x$ for large n . Cover all $d \leq n$ by:

- (1) the main family $\mathcal{G}_{\epsilon, C}(n)$;

(2) bundles $\mathcal{B}_q(n)$ for

$$q \in \mathcal{Q}_n = \{q : Y \leq q \leq Y^2\} \cup \{p \text{ prime} : Y \leq p < \epsilon x\};$$

(3) the individual $A_d(n)$ with $d < Y$;

(4) for each $1 \leq a < Y$, the tail $\{A_{ap}(n) : p > Cx \text{ prime}, ap \leq n\}$.

To verify the cover, suppose d has no prime factor in $[Y, Cx]$. At most one prime factor, including multiplicity, exceeds Cx , since $C > 1$ and $d \leq n$. Thus $d = s$ or $d = sp$, where all prime factors of s are below Y and $p > Cx$ is prime. If $s \geq Y$, multiply its prime factors until the product first reaches Y ; the resulting divisor lies in $[Y, Y^2)$. If $s < Y$, type (3) or (4) applies.

By [Corollary 3.2](#),

$$\begin{aligned} M_Q &\leq (n + \epsilon x + 1) \sum_{q \in \mathcal{Q}_n} (1 + K_0 n^{3/5} q^{-8/5}) \\ &\leq (n + \epsilon x + 1) (\pi(\epsilon x) + Y^2 + O(n^{3/5} Y^{-3/5})). \end{aligned}$$

Here $Y^2 = O(n^{2/5})$ and $n^{3/5} Y^{-3/5} = O(n^{12/25})$, so after multiplication their contributions are $o(T_n)$. The prime number theorem gives

$$M_Q \leq (2\epsilon + o(1))T_n. \quad (\text{A.9})$$

Type (3) costs at most $nY = o(T_n)$.

For type (4), put $P = Cx$ and $N_a = \lfloor n/a \rfloor$. Since $C > 2$, $P \geq 2\sqrt{N_a}$. Apply [Lemma 7.2](#) to the $A_p(N_a)$ and then [Lemma 7.1](#), using $A_{ap}(n) = aA_p(\lfloor n/a \rfloor)$. The a -th tail costs at most

$$B \left(\frac{N_a^2}{P \log P} + N_a \log(2N_a) \right) + n + a.$$

Summing $a < Y$, using $\sum a^{-2} < 2$ and $\sum_{a < Y} a^{-1} = O(\log Y)$, gives

$$M_{\text{tail}} = O \left(\frac{n^2}{P \log P} \right) + O(n \log^2 n + nY + Y^2) \leq \left(\frac{B_0}{C} + o(1) \right) T_n. \quad (\text{A.10})$$

Concatenate all packings and delete duplicate labeled copies. Combining [Proposition 6.1](#), (A.9), and (A.10),

$$\limsup_{n \rightarrow \infty} \frac{m(n)}{T_n} \leq \left(\frac{4}{3} - 2\epsilon + \frac{2}{3}\epsilon^3 \right) + 2\epsilon + \frac{B_0}{C},$$

which is (8.1). □

Theorem 1.2 (Main theorem). *As $n \rightarrow \infty$,*

$$m(n) = \left(\frac{4}{3} + o(1) \right) \frac{n^{3/2}}{\log n}.$$

Proof. [Proposition 2.2](#) gives the lower bound. For the upper bound, [Proposition 8.1](#) holds for every fixed $0 < \epsilon < 1$ and $C > 2$. Let $\epsilon \downarrow 0$ and then $C \rightarrow \infty$ to obtain

$$\limsup_{n \rightarrow \infty} \frac{m(n)}{T_n} \leq \frac{4}{3}.$$

Together with the lower bound this proves the theorem. □

B Report of the LLM referee (verbatim)

The following is the unedited report of the adversarial referee agent (claude-opus-5, 2026-09-17) on the *original* writeup. Its step, section and equation numbers refer to that writeup, not to this note. The scripts it mentions are in `verification_astra/scripts/2603.02786__00/`. Treat it as machine output, not as an authority.

Target claim. $m(n) = m_{[n]}(n) \sim \frac{4}{3} \cdot \frac{n^{3/2}}{\ln n}$, i.e. the leading-constant conjecture of Alon, Dębski, Grytczuk and Przybyło (arXiv:2603.02786, Theorem 1 gives $4/3$ and $5/3$) for the *full* difference set $\{A_d = D_d \cap [n] : d \in [n]\}$.

I went in expecting to find the error. I did not find one. What I found instead is a **priority problem**: the hard, constant-determining half of the argument was posted to arXiv by an independent group nine days before this attack ran.

B.1 Interpretation

Checked against `prompt.md` and against the source paper’s own notation (via arXiv:2609.07487 §1, which restates it verbatim).

- $A_d(n) = \{d, 2d, \dots, \lfloor n/d \rfloor d\}$, integer shifts, packing length = number of integer points of the host interval. The writeup uses exactly this and notes (correctly) that cardinality vs. diameter shifts everything by $O(1)$.
- $\log$ is natural throughout (the writeup uses $\log n = 2 \log x$ together with the PNT), matching the conjecture’s $\ln n$.
- The theorem proved is the **full** conjecture $m_{[n]}(n) \sim \frac{4}{3} T_n$, not a prime-restricted or otherwise weakened surrogate. Section 7 explicitly covers every index $d \leq n$, and I verified that cover exhaustively by computer (below). There is no interpretation gaming: an author of the original problem would consider this the intended statement.
- The claim is *asymptotic only*; nothing about a rate is claimed, and the writeup says so. That matches the conjecture’s “ $\sim$ ”.

The retry prompt contained the earlier gpt-5.6-sol1 attempt (verdict `partial`, prime subfamily only, upper constant $(8\sqrt{2} - 7)/3$). The new writeup does **not** inherit it uncritically: it reproves the separation lemma from scratch in §1 (I checked that proof independently) and it explicitly discards the path/triple condition (3.2) that stalled the earlier attempt. It does not reproduce the earlier attempt’s limitation.

B.2 Step-by-step findings

step	label	note
1. §1 separation bound (1.1): disjoint $t + A_p$, $u + A_q \Rightarrow u - t > r_p p - p q$	VALID	Re-derived. $\delta + j q \equiv 0 \pmod{p}$ has $j \in [1, p]$; $i = (\delta + j q)/p$ satisfies $1 \leq i \leq r_p$ because $\delta \leq r_p p - p q$; $j \leq p \leq \lfloor n/q \rfloor$ needs $p q \leq n$, which holds since $p < q \leq \sqrt{n}$. This is the source paper’s lemma, correctly restated.
2. §1 telescoping $\rightarrow$ (1.2), $\sum p_i p_{i+1} \leq \sum_{p \leq x} p^2$	VALID	Rounding losses $(\pi(x) - 1)x = O(n/\log n) = O(n)$, as claimed.
3. §1 PNT $\rightarrow$ (1.3) constant $4/3$	VALID	$n\pi(x) \sim 2T_n$, $\sum_{p \leq x} p^2 \sim \frac{x^3}{3 \log x} = \frac{2}{3} T_n$; difference $4/3$. Checked numerically (very slow convergence, see Computational check).
4. Lemma 2.1 degree bound: neighbours of $a > \sqrt{N}$ are divisors of ka , $k \leq N/a$	VALID	$\text{lcm}(a, b) \leq N$ forces $\text{lcm} = ka$ with $k < \sqrt{N}$ and $b \mid ka$; degree $\leq \sqrt{N} \max_{t \leq N} \tau(t)$. Verified computationally up to $N = 6 \cdot 10^4$.
5. Lemma 2.1 divisor bound $\tau(t) = O(t^{1/10})$ and $R(N) \leq K_0 N^{3/5}$	GAP (cosmetic)	The inequality chain $e + 1 \leq 2^e \leq p^{e/10}$ is correct for $p \geq 2^{10} = 1024$ and the small-prime factor is finite, so the bound is true. But K_0 is astronomically large ($\prod_{p < 1024} \sup_e (e + 1) p^{-e/10}$, order $10^8 - 10^{10}$); the writeup calls this “elementary” and moves on. Harmless — only $R = o(p)$ is ever used — but the one-line justification is glib. Measured greedy values are far below the bound ($R(N)/N^{3/5} = 0.96, 0.73, 0.59, 0.50, 0.44$ for $N = 200 \dots 6 \cdot 10^4$). $\lfloor \lfloor n/q \rfloor / a \rfloor = \lfloor n/(qa) \rfloor$.
6. (2.2) $A_{qa}(n) = q A_a(\lfloor n/q \rfloor)$	VALID	Colours get distinct residues mod q ;
7. Corollary 2.2 bundle packing in $(n + q) \lfloor R/q \rfloor$	VALID	same-colour sets stay disjoint under a common shift. Verified by code for $(n, q) = (20000, 37)$ and $(20000, 211)$: no collision, $\text{span} \leq \text{bound}$.

step	label	note
8. (2.4) $R(\lfloor n/p \rfloor) = O(n^{3/10}) = o(p)$ for $p \asymp \sqrt{n}$	VALID	$(n/p)^{3/5} = O(x^{3/5}) = O(n^{3/10})$.
9. (3.3) CRT phase formula	VALID	$Z_{ij} = \overline{p_j^{(p_i)}} r_i / p_i + \overline{p_i^{(p_j)}} r_j / p_j$; checked against the standard CRT representative.
10. (3.4) expectation factorises to $\prod_i \mathbf{1}[c_i \equiv 0]$	VALID	$\sum_{i < j} h_{ij} Z_{ij} = \sum_i \frac{r_i}{p_i} \sum_{j \neq i} h_{ij} \overline{p_j^{(p_i)}}$ using $h_{ji} = h_{ij}$; independence of the r_i then gives the product.
11. Finite Fourier criterion (3.5)	GAP (cosmetic)	The argument is right ($\widehat{F}(0) = \mu^E$, middle frequencies killed, tail $< \mu^E/2$ uniformly in the centres). But f is only said to be “supported inside an arc of length η ” while J_{ij} is an <i>open</i> arc of length η ; strictly one must take $\text{supp } f$ inside an arc of length $< \eta$ (e.g. $\eta/2$) to guarantee $Z_{ij} \in J_{ij}$. One-line repair.
12. Lemma 3.1 bad-tuple count $O_{k,a,b,\eta}(M^{k-1})$	VALID	For $p_i > H$ and $h_{ij} \neq 0$: $c \equiv 0$ gives no solution ($\overline{p_j^{(p_i)}} \neq 0$); otherwise p_j is pinned to one class mod $p_i \geq aX$, hence $O_{a,b}(1)$ integers in $[aX, bX]$. Summing over the $(2H+1)^{\binom{k}{2}}$ vectors h gives the claim with an M -independent constant, so a good block exists once M exceeds a constant; iterate.
12b. (unremarked subtlety, not an error)	NOTE	Every prime triple $p_0 < p_1 < p_2$ is bad at frequency $\ h\ _\infty = p_2 - p_0$: with $d_1 = p_1 - p_0$, $d_2 = p_2 - p_0$, $d_3 = p_2 - p_1$, the vector $h = (h_{01}, h_{02}, h_{12}) = (d_1, -d_2, d_3)$ satisfies all three congruences identically. I verified this on 300 random triples. So Lemma 3.1 is true <i>only</i> because $H = H(k, \eta)$ is fixed while the primes spread over $[aX, bX]$; the counting argument covers this correctly (close triples are exactly the $O(M^{k-1})$ bad ones), but the writeup never points out that this is the sharp edge of the lemma. My first packing simulation failed precisely because I picked three nearly-consecutive primes — a bad block.
13. Lemma 4.1 Bohr set $\geq c(d, \rho)q$	VALID	Pigeonhole on M^d boxes, then difference with a representative; $c = M^{-d}$, $M > 1/\rho$. Sizes matched the $(2\rho)^d q$ heuristic in the simulation.
14. §5.1 inequality (5.2)	VALID	$W - g \leq n + Cx + 2 - n(h(u) + \nu) \leq (\min(u^2, 1) - \nu/2)n$ for large n , and $\min(u^2, 1)n \leq p_i p_j$ since $p_i, p_j \geq ux$ (case $u > 1$: $p_i p_j > u^2 n > n$). Both inequalities re-derived and asserted inside the simulation.
15. §5.1 arc geometry: complement $\geq \nu/(2C^2) = 50\rho$, buffer 4ρ , drift $< 2\rho$	VALID	$p_i p_j \leq C^2 n$ gives the complement bound; 50ρ comfortably accommodates a ρ -arc with 4ρ clearance; (5.6) drifts each phase by $< 2\rho$ by (5.5).
16. §5.1 disjointness of the packed block	VALID	Within a bundle: same colour disjoint by the unshifted colouring, different colours by distinct residues mod p_i . Across bundles: any common point lies in $I_i \cap I_j$ and has phase $Z_{ij}(\alpha_{i,c}, \alpha_{j,c'})$, which is kept outside the projection of $I_i \cap I_j$. The argument only ever uses residues modulo the two <i>primes</i> , which is exactly why non-coprimality of the composite indices is harmless. Verified end-to-end by brute force (see below).
17. (5.7) block span $(k-1)g + W$; (5.8) summation	VALID	$\lceil M_j/k \rceil$ blocks per bin, blocks concatenated into disjoint intervals.

step	label	note
18. §5.2 PNT + mesh $\rightarrow 0 + \nu \downarrow 0 + k \rightarrow \infty$ $\rightarrow (5.1), (5.9)$	VALID	$nx/\log x = 2T_n$; h decreasing so $\sum(u_j - u_{j-1})h(u_{j-1})$ is an upper Riemann sum. $2 \int_{\epsilon}^1 (1 - u^2)$ $du = \frac{4}{3} - 2\epsilon + \frac{2}{3}\epsilon^3$ (checked symbolically). Order of limits is legitimate: every parameter is fixed before $n \rightarrow \infty$ and the lim sup does not depend on them.
19. Lemma 6.1 folding	VALID	Strip map $B \mapsto a(B - jH) + j$; same strip stays disjoint (affine injection), different strips separated mod a ; length $\leq aH + aN \leq M + aN + a$. 200 random instances verified.
20. Lemma 6.2 large-prime greedy	VALID	$K = \lfloor Q^2/2N \rfloor \geq Q^2/4N$; a group of $\leq K$ progressions occupies $< KN/Q \leq Q/2 < p$ points, each forbidding one residue. Dyadic sum $\sum 1/(Q \log Q) \leq 2/(P \log P)$; $O(\log 2N)$ bins. Greedy verified to always succeed for $(N, P) = (5000, 200)$ and $(20000, 400)$. At most one prime factor $> Cx$ since $C^2 x^2 > n$; if the $< Y$ -smooth part is $\geq Y$ it has a divisor in $[Y, Y^2]$. Verified exhaustively for $n = 10^4, 10^5, 10^6$: zero uncovered indices.
22. §7.1 cost $(2\epsilon + o(1))T_n$	VALID	$\sum_{q \geq Y} q^{-8/5} = O(Y^{-3/5})$ gives $O(n^{12/25})$ extra batches, so $n \cdot n^{12/25} = n^{37/25} = n^{1.48} = o(T_n)$; $nY^2 = n^{1.4} = o(T_n)$; main term $n\pi(\epsilon x) \sim 2\epsilon T_n$. Arithmetic re-derived. $P = Cx \geq 2\sqrt{N_a}$ needs $C > 2$ (it is assumed); $\sum_a a^{-2} < 2$, $\sum_a a^{-1} = O(\log Y)$; $n^2/(Cx \log(Cx)) \leq 2T_n/C$. $(\frac{4}{3} - 2\epsilon + \frac{2}{3}\epsilon^3) + 2\epsilon + B_0/C =$ $\frac{4}{3} + \frac{2}{3}\epsilon^3 + B_0/C$; the -2ϵ from the truncated integral is exactly cancelled by the type-2 bundles. $\epsilon \downarrow 0, C \rightarrow \infty$.
23. §7.2 tail $\leq (B_0/C + o(1))T_n$	VALID	
24. §8 combination	VALID	

No step is labelled ERROR.

B.3 Reference check

The writeup invokes **no external theorem at all** except the prime number theorem and elementary divisor/CRT facts, all of which I re-derived. So there is no misquoted citation to catch. That, however, is also the problem:

Priority (step 2b). I found an independent preprint that already contains the constant-determining half of this argument.

- **arXiv:2609.07487**, *Asymptotically optimal packings of arithmetic progressions with prime differences*, Jianfeng Hou, Siyue Liu, Hongbin Zhao (Center for Discrete Mathematics, Fuzhou University), submitted **7 Sep 2026** — nine days before this attack ran (16 Sep 2026). Theorem 1.1: $m_{\mathcal{P}(\sqrt{n})}(n) = (\frac{4}{3} + o(1)) \frac{n^{3/2}}{\ln n}$. Their method: “cyclic phase selection” (Lemma 2.2 there is the same interval-vs-CRT-period avoidance the writeup uses) plus §3 “**Regular prime blocks**”, which partitions primes in $(P, (1+\eta)P]$ into good transversals of fixed size $s \geq 3$ by exactly the writeup’s criterion — no nonzero h with $\|h\|_2 < R$ satisfying $\sum h_i p_i^{-1} \equiv 0 \pmod{r}$. This is the writeup’s Lemma 3.1 in different clothing (they get simultaneity from Banaszczyk’s transference inequality where the writeup uses a Weyl-sum/bump-function argument).
- **This is not self-contamination (step 2c).** I opened the actual arXiv abstract page and the full HTML: real authors with institutional e-mail addresses, a Chinese NKRD grant number, a 12-item bibliography (Alon–Bóna–Spencer, Banaszczyk, Coven–Meyerowitz, Ford, Green–Ruzsa, Katz–Tao, Montgomery–Vaughan, ...), and an independent Semantic Scholar record (CorpusId 291870588, 0 citations). Nothing traces back to this repository.
- **Crucially, it does not cover the target.** Their §6 Concluding remarks state plainly: “A direct extension of the present method to arbitrary differences would have to account for nontrivial common divisors. For example, the differences $p, 2p, \dots$ are not pairwise coprime, so the modular inverses used in the current phase parameterization are unavailable. ... obtaining the same leading constants for the full difference set $[n]$ remains a further problem.” That is precisely the obstacle the writeup’s §2 (bundle colouring into $o(p)$ residue classes) + §4 (Bohr set clustering) + §7 (cover) remove. Nothing cites 2609.07487 yet.

So: **the claimed theorem is not in the literature; its load-bearing prime-range lemma is.** The writeup nowhere acknowledges this (it could not have — it cites nothing), and a referee would require the overlap to be stated.

Two neighbouring facts worth recording for the sibling reviews:

- **arXiv:2607.06113** (Mao, Wang, Wei, Yang, 7 Jul 2026) proves $M_k(n) = (1 + o(1))nk$ for fixed n , settling Conjecture 7 of 2603.02786. The sibling target 2603.02786__04 claims exactly “ $M_k(n) = nk + o(k)$ for every fixed n ” — that looks like the same theorem and should be checked for ALREADY_KNOWN first.
- The source paper’s own $5/3$ upper bound corresponds to capturing only half the pairwise overlap saving: $2 \int_0^1 (1 - u^2/2) du = 5/3$, against $2 \int_0^1 (1 - u^2) du = 4/3$ for the full saving. I verified both integrals. This confirms that the writeup is attacking the right mechanism rather than an artefact.

B.4 Computational check

Scripts in `verification_astra/scripts/2603.02786__00/` (python3, sympy/numpy).

1. `check_constants.py` — symbolic: $2 \int_0^1 (1 - u^2) du = 4/3$, $2 \int_0^1 (1 - u^2/2) du = 5/3$, $2 \int_\epsilon^1 (1 - u^2) du = \frac{4}{3} - 2\epsilon + \frac{2}{3}\epsilon^3$. Numeric PNT sums: $(n\pi(\sqrt{n}) - \sum_{p \leq \sqrt{n}} p^2)/T_n = 1.639, 1.605, 1.572, 1.545, \mathbf{1.522}$ for $n = 10^6 \dots 10^{10}$. Still 14% above $4/3$ at $n = 10^{10}$ — the writeup’s caveat about the absence of a usable rate is an understatement, not a formality.
2. `check_lemma21.py` — max degree of the lcm-intersection graph, greedy chromatic number, and $\max_{t \leq N} \tau(t)/t^{1/10}$. $R(N)/N^{3/5} = 0.96, 0.73, 0.59, 0.50, 0.44$ for $N = 200, 10^3, 5 \cdot 10^3, 2 \cdot 10^4, 6 \cdot 10^4$: Lemma 2.1 holds with room to spare.
3. `universal_relation.py` — verified on 300 random triples that $h = (p_1 - p_0, -(p_2 - p_0), p_2 - p_1)$ always satisfies all three congruences of (3.4); and that [6007, 6011, 6029] is bad at $\|h\|_\infty = 11$ (witness (2, -11, 9)) while [6007, 6373, 6997] and [6113, 6521, 6959] are good up to $\|h\|_\infty \leq 12$.
4. `check_lemma31.py` — 0/40 random triples from [6000, 7000] bad at $H = 3$. Exact realisability counts for the bad triple [6007, 6011, 6029] with $\rho = 0.03$: for six random arc triples the number of admissible (r_0, r_1, r_2) was **0**, 12.4M, 11.1M, 17.1M, 15.3M, 11.3M against the equidistribution prediction 5.9M — i.e. a bad block really can make (3.2) unsatisfiable, confirming that Lemma 3.1’s genericity hypothesis is doing essential work.
5. `check_block_packing.py` — **the decisive test**. Full literal instantiation of §5.1 at $n = 10^8$, $x = 10^4$, block {15013, 15511, 15991} (verified CRT-generic up to $\|h\|_\infty \leq 25$), $u = 1.5$, $\nu = 0.10$, $C = 2$, $\rho = 0.06$, $k = 3$, so that **all three host windows overlap**: host intervals, forbidden arcs (overlaps $9.00 \cdot 10^7, 8.00 \cdot 10^7, 9.00 \cdot 10^7$ against periods $2.33 \cdot 10^8, 2.40 \cdot 10^8, 2.48 \cdot 10^8$; forbidden fractions 0.387, 0.333, 0.363, complements $\geq 0.613 \geq 9\rho$), Lemma-3.1 residues $r = (9243, 2582, 11434)$, Lemma-2.1 colourings $(R(N_i) = 112, 110, 108$ for $N_i = 6660, 6447, 6253)$, Lemma-4.1 Bohr sets (sizes 213, 217, 233 — enough), then every progression of every bundle expanded. Result: **172 850 points, all pairwise distinct**, occupying a span of 119 973 288 $\leq (k-1)g + W = 120\,020\,001$. The composite-indexed progressions are included, and no collision occurs. Assertions for both halves of (5.2) and for the 9ρ complement condition pass.
6. `check_sec6.7.py` — Corollary 2.2 packings collision-free and within bound; Lemma 6.1 folding on 200 random instances (disjointness + $M + aN + a$); Lemma 6.2 greedy always finds a free residue for $(N, P) = (5000, 200), (20000, 400)$; and the §7 cover verified exhaustively for $n = 10^4, 10^5, 10^6$ — **0 uncovered indices**.

What the computation cannot test is the limiting regime itself ($\nu \downarrow 0$, $k \rightarrow \infty$, mesh $\rightarrow 0$, $\epsilon \downarrow 0$, $C \rightarrow \infty$): reaching ν small enough to see the constant $4/3$ needs n of order 10^{11} or more. Those limits I checked only by hand, and they are the standard part of the argument.

B.5 Caveats

- **Priority**. The prime-range half — Lemma 3.1 and the whole of §5, which is where the constant $4/3$ is actually produced — duplicates Theorem 1.1 of arXiv:2609.07487 (7 Sep 2026). Genuinely new here is only §2 + §4 + §6 + §7: the reduction of a whole bundle $\{A_{ap}\}$ to $o(p)$ residue classes and their clustering inside one Bohr set. That is a real idea and it answers the exact question Hou-Liu-Zhao left open — but any write-up must say so.
- **Wholly non-effective**. K_0 (divisor bound), $H(k, \eta)$ (Fourier cutoff), $L(k, a, b, \eta)$ (exceptional primes), $c(d, \rho) = M^{-d}$ (Bohr density) are all unquantified and, where one can estimate them, enormous. Combined with the measured convergence (1.52 at $n = 10^{10}$), the result is qualitative in the strongest sense.
- **Smooth-bump support** (step 11) must be shrunk below η for the open-arc conclusion; trivial.
- $\nu < \epsilon^2/4$ is imposed in §5.1 but never actually invoked; it is only needed to keep $W - g$ and the complement positive, which (5.2) already gives.
- **Lemma 3.1’s genericity is fragile in a way the text hides** (step 12b): every triple is bad at frequency $\sim p_{\max} - p_{\min}$, so the lemma lives entirely on the fixed- H /growing- X separation. The proof handles it; the exposition does not flag it, and a reader could easily believe good blocks are the rule for trivial reasons.
- **Deletion of duplicates** (indices with two prime factors in $[\epsilon x, Cx]$) is asserted in one sentence; it is correct (removing sets from a packing keeps it a packing) but is the only place where the “pack labelled copies” device is used.
- The writeup contains **no bibliography**, so “references_ok” above means “no citation is misused”, not “the literature was consulted”.

B.6 Is arXiv:2603.02786 easy, or is one trick being reapplied?

Asked because three of its sub-problems were claimed in this batch and a fourth was confirmed earlier. My reading: **mostly genuinely tractable, with one shared mechanism**.

The paper is six months old and its conjecture list is being picked off by independent human groups at speed — Conjecture 7 by Mao-Wang-Wei-Yang in July (arXiv:2607.06113), Conjectures 3 and 4 by Hou-Liu-Zhao in September (arXiv:2609.07487). These are open problems at the frontier of routine technique, not deep ones; the authors themselves only said “additional ideas” were needed. So a strong model getting traction on several of them at once is not suspicious per se.

There is a recurring trick, though, and it is the CRT one: control the residue class (phase) of each progression modulo its own difference, and use coprimality of distinct prime differences to make the intersection pattern of two progressions depend only on one phase difference mod pq . __03 (confirmed earlier) uses it in lower-bound form (cores with distinct phases mod gcd); __00 uses it in upper-bound form. That is the paper’s native technique, so its recurrence is expected rather than evidence of a template being over-applied. What I would *not* assume from this target is that the siblings are equally sound: __04’s stated conclusion ($M_k(n) = nk + o(k)$ for fixed n) appears to be literally the published theorem of arXiv:2607.06113, and __01’s own verdict block admits novelty was never checked. Both deserve a priority check before a correctness check.

B.7 Referee summary

I attacked this writeup for a full pass and could not break it. Every one of the 24 numbered steps re-derives correctly; the two defects I found (the bump function's support, the glib divisor-bound constant) are one-line repairs, and the one real subtlety — that *every* prime triple satisfies a CRT relation, so Lemma 3.1 depends entirely on the fixed frequency cutoff H against growing prime gaps — is handled correctly by the bad-tuple count even though the text never draws attention to it. I instantiated the whole §5.1 block construction literally at $n = 10^8$ with three mutually overlapping host windows and all composite-indexed progressions present, and it produced 172 850 pairwise-distinct points inside the claimed span; Corollary 2.2, Lemmas 6.1 and 6.2, and the §7 cover all pass independent tests. The mechanism that answers the conjecture — pushing a whole bundle $\{A_{ap}(n)\}_a$ into $o(p)$ residue classes that all sit inside one Bohr set, so that cross-bundle disjointness is decided purely by the coprime pair (p_i, p_j) — is correct and is exactly the device that Hou–Liu–Zhao name as the missing ingredient for general differences. What sinks the “would_publish: true” claim is not an error but the calendar: arXiv:2609.07487, posted nine days before the attack, already proves the prime-difference case $m_{\mathcal{P}(\sqrt{n})}(n) \sim \frac{4}{3}n^{3/2}/\ln n$ by the same regular-prime-block route, so the genuinely new content is the composite extension, not the constant. I record MINOR_GAPS rather than CONFIRMED because of the two cosmetic repairs and because the limiting regime that actually delivers $4/3$ is beyond computational reach (at $n = 10^{10}$ the lower-bound expression is still 1.52), and medium confidence because a claim of this size deserves a second human pass on §5.2's interchange of limits.

References

- [1] N. Alon, M. Dębski, J. Grytczuk and J. Przybyło, Packing arithmetic progressions, [arXiv:2603.02786](#) (3 March 2026).
- [2] J. Hou, S. Liu and H. Zhao, Asymptotically optimal packings of arithmetic progressions with prime differences, [arXiv:2609.07487](#) (7 September 2026).
- [3] Y. Mao, Z. Wang, M. Wei and G. Yang, Solution to a conjecture of Alon, Dębski, Grytczuk and Przybyło on fixed-cardinality arithmetic progressions, [arXiv:2607.06113](#) (7 July 2026).
- [4] G. H. Hardy and E. M. Wright, *An Introduction to the Theory of Numbers*, 6th ed., Oxford University Press, 2008.